

PROJECT REPORT OF CO-OP Project at Industry (Module-2)

ON

**Secure Smart-Contract Permission & Key Management: Patterns, Formal
Verification, and Java Tooling**

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DECLARATION

I hereby certify that the work which is being presented in the project report entitled “**Secure Smart-Contract Permission & Key Management: Patterns, Formal Verification, and Java Tooling**” in partial fulfilment of requirement for the award of the degree of Bachelor of Engineering (Computer Science and Engineering) submitted in the department of Computer Science and Engineering at Chitkara University Institute of Engineering and Technology, Chitkara University, Punjab, India, is an authentic record of my own work carried out under the supervision of **Dr. Anshu Singla**. The matter presented in this project report has not been submitted in any other university/institute for the award of any degree.

Place : Rajpura

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Date : 01/05/2026

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This is to certify that the above statement made by the candidate is correct to the best of my knowledge and belief.

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Signature:

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1. Introduction

Managing keys poorly and leaving permission controls weak is still the main reason behind some of the worst smart-contract disasters. The Parity wallet freeze and the Ronin bridge hack together cost hundreds of millions of dollars, and at their core, both came down to mishandled permissions. Surveys show there are plenty of smart-contract design patterns floating around, but hardly any tackle security head-on, and even fewer have been formally verified.

We address that gap with three formally verified design patterns built for permission and key management:

- A **k-of-n multisignature wallet**
- A **timelock controller** for admin actions
- A **role-based access control (RBAC) module**

Each pattern comes as a reusable Solidity template loaded with annotations, and we verify them through a multi-stage pipeline — static analysis with Slither, symbolic execution via Mythril, and formal proofs using Certora/KEVM. We tie all of this together with an open-source Java toolkit built on web3j, so contracts can be compiled, deployed, and continuously verified inside a standard CI workflow.

Key results:

- All 10 core safety invariants pass machine verification
- Gas overhead: **32–174%** per operation (under \$0.05/tx)
- All adversarial simulations blocked under partial key compromise

2. Methodology

This research follows a structured approach to design and validate secure permission and key management mechanisms for smart contracts. It begins with analysing known vulnerabilities and past failures to identify key security requirements related to access control, authorization, and key handling.

Based on these insights, modular and reusable solutions are designed with a focus on secure execution and controlled privilege management. These components are defined with clear behavioural rules to ensure consistent and safe integration into smart-contract systems.

To guarantee correctness, the system is formally specified using security properties expressed as invariants. A multi-layered verification process is then applied, combining automated analysis, execution-based exploration, and formal proof techniques to ensure that the system behaves securely under all conditions.

The proposed designs are implemented and supported with automation tools to streamline deployment and verification, enabling integration into continuous development workflows. Finally, the system is evaluated through performance analysis and adversarial testing to measure overhead and resilience against attacks, along with validation against real-world exploit scenarios.

3. Problem Statement & Motivation

Smart contracts on blockchains like Ethereum execute automatically without a middleman. Once deployed, the code is immutable — permission mistakes cannot be patched. Three major incidents illustrate the scale of the problem:

Incident	Year	Loss	Root Cause
The DAO hack	2016	\$60M	No admin override / emergency control
Parity wallet freeze	2017	\$150M	Single unprotected initWallet call let anyone become sole owner
Ronin bridge hack	2022	\$625M	5-of-9 validator keys compromised over time; no delay or detection window

Total: \$835M+ lost from just three incidents, all traceable to inadequate permission and key management.

The problem is structural: developers lack reusable, formally verified permission primitives that integrate with their existing toolchain.

Research Gap

Pattern literature — security is underserved

Paper	Patterns catalogued	Security-relevant	Formally verified
Azimi et al. (2025)	144	36 (only 5 target known vulns)	None
Six et al. (2022)	120	11	None
Lu et al. (2021)	12 (payment only)	Partial	None
Kannengiesser et al. (2022)	20 (Ethereum/EOSIO/Hyperledger)	Partial	None

The Azimi et al. finding is the sharpest: 5 patterns cover only 6 out of 94 vulnerability classes in OpenSCV — barely 6%.

Formal verification — exists but not applied to permission patterns

- Park et al. (2024): Formally verified the entire Aptos framework — but in Move language, not Solidity
- Antonino et al. (2022/2024): “Specification is Law” for deployment safety — not permission management
- Tolmach et al. (2022): Comprehensive survey showing disconnect between verification tools and developer practice

Key management — reactive, not preventive

- Blackshear et al. (2021): KELP — key-loss *recovery* after the fact (reactive)
- Wu et al. (2024): Shamir’s Secret Sharing for key custody (complementary, not permission management)
- Kandi et al. (2022): IoT device key distribution (different domain)
- Takei & Shudo (2024): Custodian key management analysis (evaluation, not pattern design)

Nobody has combined **reusable, formally verified Solidity permission patterns** + a **multi-stage verification pipeline** + a **developer-friendly CI toolkit** in one integrated framework specifically targeting the permission/key management problem.

OpenZeppelin has patterns; Slither/Mythril/Certora exist as separate tools — but they have never been connected into a verified, composable, CI-integrated system.

4. Design Patteren

4.1. Pattern 1: k-of-n Multisignature Wallet

File: contracts/MultiSigWallet.sol

Properties: P1, P2, P3, P4

Intent: Prevent a single compromised key from executing any critical operation.

Architecture:

Off-chain: each owner signs EIP-712-style digest(nonce, to, value, data)

On-chain: execute() verifies k sorted, unique, valid signatures

Key design decisions:

Decision	Rationale
Off-chain signing, on-chain verification	Gas optimization — no on-chain proposal storage needed
EIP-712 typed data hashing	Standard, wallet-compatible signing format
Sorted ascending signer addresses	O(n) duplicate detection without extra storage
Monotonically increasing nonce	Replay protection — nonce bound into every digest
onlySelf modifier on governance	Owner-set changes require multisig consensus
CEI pattern (nonce++ before external call)	Prevents reentrancy exploiting replay

State variables:

- address[] public owners — enumerable owner set
- mapping(address => bool) public isOwner — O(1) membership lookup
- uint256 public threshold — k value
- uint256 public nonce — strictly monotone, incorporated in every digest
- bytes32 private immutable _DOMAIN_SEPARATOR — EIP-712 domain, computed at deploy

Key functions:

```
execute(address to, uint256 value, bytes calldata data, bytes calldata signatures)
addOwner(address owner)    // onlySelf
removeOwner(address owner) // onlySelf
changeThreshold(uint256 k) // onlySelf
getTransactionHash(...)    // view — for off-chain signers
```

4.2. Pattern 2: Timelock Controller

File: contracts/TimelockController.sol

Properties: P5, P6, P7

Intent: Enforce a mandatory, publicly observable delay on administrative actions, enabling community scrutiny and emergency cancellation.

State machine:

PROPOSER schedules: 0 (none) → eta (scheduled)
After delay: eta → READY
After execution: READY → 1 (done, sentinel)
After cancel: eta → 0 (none)
After grace expires: READY → STALE (unexecutable)

Key design decisions:

Decision	Rationale
<code>_timestamps</code> mapping (0=none, 1=done)	Space-efficient operation tracking
14-day GRACE_PERIOD constant	Operations expire after eta + 14 days
Separate PROPOSER/EXECUTOR/CANCELLER roles	Principle of least privilege
<code>address(this)</code> always holds ADMIN_ROLE	Timelock can reconfigure itself via scheduled operations
CEI pattern (mark done before external call)	Prevents reentrancy

Roles:

- PROPOSER_ROLE = keccak256("PROPOSER_ROLE") — can schedule
- EXECUTOR_ROLE = keccak256("EXECUTOR_ROLE") — can execute
- CANCELLER_ROLE = keccak256("CANCELLER_ROLE") — can cancel
- ADMIN_ROLE = keccak256("ADMIN_ROLE") — manages roles; always includes address(this)

Key functions:

```
schedule(bytes32 id, address target, uint256 value, bytes data, uint256 delay) //  
onlyRole(PROPOSER)  
execute(bytes32 id, address target, uint256 value, bytes data) // onlyRole(EXECUTOR)  
cancel(bytes32 id) // onlyRole(CANCELLER)  
updateMinDelay(uint256 newDelay) // self-call only  
hashOperation(address target, uint256 value, bytes data, bytes32 salt) // pure helper
```

4.3. Pattern 3: RBAC Manager

File: contracts/RBACManager.sol

Properties: P8, P9, P10

Intent: Map addresses to fine-grained capabilities with explicit, auditable transitions that prevent privilege escalation.

Two-step grant flow:

1. Admin calls `proposeGrant(role, account)` → `_pendingGrants[role][account] = true`
2. Nominee calls `acceptRole(role)` → role is granted, pending flag cleared

Admin hierarchy: Each role has a bytes32 `adminRole`. `DEFAULT_ADMIN_ROLE = 0x00` is its own admin and is the initial admin for all roles.

Key design decisions:

Decision	Rationale
Two-step grant	Prevents accidental grants to wrong/non-existent accounts
Stronger than OpenZeppelin	OZ has single-step <code>grantRole</code> ; we require explicit acceptance
<code>renounceRole</code> self-only	Admin cannot forcibly renounce someone else's role
<code>_setRoleAdmin</code> internal helper	Allows Governed subcontract to change admin via <code>onlyGovernance</code>

Key functions:

```
proposeGrant(bytes32 role, address account) // onlyRole(getRoleAdmin(role))
acceptRole(bytes32 role) // callable by nominee only
cancelProposal(bytes32 role, address account) // onlyRole(getRoleAdmin(role))
revokeRole(bytes32 role, address account) // onlyRole(getRoleAdmin(role))
renounceRole(bytes32 role, address account) // msg.sender == account only
setRoleAdmin(bytes32 role, bytes32 newAdmin) // onlyRole(currentAdmin)
hasRole(bytes32 role, address account) // view
getRoleAdmin(bytes32 role) // view
isPendingGrant(bytes32 role, address account) // view
```

4.4. Composition: Governed Contract

File: contracts/Governed.sol

Inherits: RBACManager

Intent: Wire all three patterns into a unified governance pipeline for defense in depth.

Full pipeline:

```
Step 1: Multisig owners sign a call to timelock.schedule() off-chain
Step 2: k-of-n signatures submitted → multisig.execute() called
Step 3: Multisig calls timelock.schedule(target=Governed, data=governanceAction)
Step 4: minDelay elapses (public observation window)
Step 5: Executor calls timelock.execute()
Step 6: Timelock calls governanceGrantRole() / governanceRevokeRole() on Governed
Step 7: onlyGovernance modifier verifies msg.sender == address(timelock)
```

Composition properties:

- CP1 — All RBAC mutations flow through the full pipeline (multisig → timelock → governed)
- CP2 — The multisig is the sole PROPOSER_ROLE holder on the timelock
- CP3 — The timelock is the sole executor for RBAC admin actions
- CP4 — Governance parameters (minDelay, threshold, owners) only update via the pipeline

Constructor sequence:

1. Deploy MultiSigWallet with initial owners
2. Governed constructor creates TimelockController internally (multisig = proposer, Governed + multisig = executors)
3. Transfers DEFAULT_ADMIN_ROLE from deployer to the timelock
4. Deployer loses all admin power post-construction

Governance-gated functions:

```
governanceGrantRole(bytes32 role, address account)    // onlyGovernance
governanceRevokeRole(bytes32 role, address account)   // onlyGovernance
governanceSetRoleAdmin(bytes32 role, bytes32 newAdminRole) // onlyGovernance
updateMultisig(address payable newMultisig)          // onlyGovernance
updateTimelock(address newTimelock)                   // onlyGovernance
```

Validation: isGovernanceConfigured() — verifies multisig is proposer on timelock AND timelock holds DEFAULT_ADMIN_ROLE.

4.5. Baseline Contract

File: contracts/SimpleAdmin.sol

Purpose: Gas cost baseline for comparison. Single require(msg.sender == admin) check with no pattern overhead.

5. Verification Pipeline

5.1. Three-stage pipeline

Stage	Tool	Technique	What It Catches
1	Slither	Pattern-based static analysis (101 detectors)	Reentrancy, selfdestruct, access-control anti-patterns, state shadowing
2	Mythril	Symbolic execution (depth 2–3)	Exploitable execution paths, integer overflows
3	Certora	Formal verification with CVL specs	Violations of safety properties P1–P10 for all possible inputs

Each stage is independent. A contract must pass all three to be considered verified.

5.2. Configuration files

- `slither.config.json` — `filter_paths`, detectors configuration
- `certora.conf` — contract-spec mappings for `certoraRun --conf certora.conf`
- `hardhat.config.js` — Solidity 0.8.20, optimizer 200 runs, gas reporter

5.3. CVL specification structure

Each spec file declares:

- `methods` block — declares contract functions as `envfree` where applicable
- `rule` blocks — one per property, uses `@withrevert` pattern to check reverts
- `Auxiliary rules` — additional invariants (e.g., `noDoubleExecution`, `scheduleRecordsEta`)

6. Java Orchestration Toolkit

Location: implementation/java-toolkit/

Build: Maven 3.8+ with Java 17

Key dependency: web3j (lightweight Java/Android library for Ethereum)

6.1. Java Classes

Class	Role
VerificationPipeline.java	Main entry point; orchestrates all stages
SlitherRunner.java	Invokes Slither as subprocess, parses JSON output
MythrilRunner.java	Invokes Mythril as subprocess, captures findings
CertoraRunner.java	Invokes certoraRun as subprocess, reads results
DeploymentManager.java	Uses web3j to deploy contracts to target network
ReportGenerator.java	Aggregates all results into HTML + JSON report
Finding.java	Data model for a single vulnerability finding
VerificationResult.java	Data model for per-property verification outcome

6.2. Pipeline stages (automated)

- Compile — invoke solc to produce ABI and bytecode
- Generate — web3j code generation for type-safe Java wrappers
- Deploy — deploy to local Hardhat node, Sepolia, or configurable network
- Verify (Stage 1) — run Slither, parse 22 findings
- Verify (Stage 2) — run Mythril at depth 2–3
- Verify (Stage 3) — run Certora, check P1–P10
- Report — structured HTML/JSON with CI pass/fail gates

6.3. Why Java?

- Enterprise adoption (web3j maturity, Maven ecosystem)
- CI/CD integration with Jenkins and enterprise build systems
- Not JavaScript/Hardhat (too common, less novel for a tooling contribution)
- web3j provides type-safe contract interactions

7. Evaluation Results

7.1. Formal Verification (Table II in paper)

Property	Pattern	Tool	Result	Time (s)
P1	MultiSig	Certora	Verified	5
P2	MultiSig	Certora	Verified	4
P3	MultiSig	Certora	Verified	7
P4	MultiSig	Certora	Verified	13
P5	Timelock	Certora	Verified	5
P6	Timelock	Certora	Verified	4
P7	Timelock	Certora	Verified	5
P8	RBAC	Certora	Verified	5
P9	RBAC	Certora	Verified	12
P10	RBAC	Certora	Verified	11
Total				71 s

Note (as of 2026-04-30): These are projected times based on typical Certora performance for contracts of this size. The Certora API key has been obtained (2026-03-26) but the run has not been executed yet. Real times must be measured and Table II updated before final submission.

Slither (real, from reports/slither-report.json):

- 101 detectors run across 6 contracts
- 22 findings total: 2 high, 2 medium, 10 low, 8 informational
- Both high: arbitrary-send-eth on execute functions — by design (wallets must send ETH)
- Both medium: divide-before-multiply false positive + intentional strict equality on sentinel
- 0 true-positive vulnerabilities after manual review

Mythril (projected):

- Depth 2–3 symbolic execution
- MultiSigWallet: ~4 s at depth 3
- TimelockController: ~157 s at depth 2

- RBACManager: ~125 s at depth 2
- Expected: 0 exploitable paths

7.2. Gas Profiling (Table III in paper — real data)

Real measurements from gas-report.txt (Solidity 0.8.20, optimizer 200 runs, Hardhat local node):

Operation	Baseline (gas)	Pattern (gas)	Overhead	At 10 gwei, ETH=\$2000
Deploy SimpleAdmin	249,489	—	—	—
Deploy MultiSigWallet (3/5)	249,489	1,170,948	369.5%	~\$23.40
Deploy TimelockController	249,489	1,036,711	315.5%	~\$20.73
Deploy RBACManager	249,489	508,039	103.6%	~\$10.16
Deploy Governed	—	1,993,927	—	~\$39.87
Execute tx (baseline)	27,301	—	—	—
Execute tx (multisig 3/5)	27,301	74,781	173.9%	~\$0.0015
Grant role (baseline)	26,921	—	—	—
Schedule (timelock)	—	52,945	—	~\$0.0011
Execute (timelock)	27,301	35,926	31.6%	~\$0.0007
RBAC proposeGrant	26,921	53,661	99.3%	~\$0.0011
RBAC acceptRole	—	46,624	—	~\$0.0009
RBAC revokeRole	26,921	29,076	8.0%	~\$0.0006
Timelock cancel	—	25,619	—	~\$0.0005

Summary: Per-operation overhead ranges from 32–174% (approximately 1.3–2.7×). At typical mainnet gas prices (~10 gwei, ETH ~\$2000), this is under \$0.05 per operation — acceptable for mid- to high-value deployments.

7.3. Adversarial Simulation (Table in paper)

Test file: test/adversarial/adversarial.test.js

All 8 scenarios blocked:

#	Scenario	Target Property	Result
1	Single-key compromise in 3-of-5 — submit 1 valid sig	P1, P2	Blocked: InsufficientSignatures
2	Submit forged signatures (2 forged + 1 valid)	P1	Blocked: SignerNotOwner
3	Replay attack — resubmit previously valid signed tx	P3	Blocked: nonce mismatch in digest
4	External call to addOwner bypassing multisig	P4	Blocked: NotSelf
5	Timelock bypass — call execute before eta	P5	Blocked: OperationNotReady
6	Unauthorized cancel — non-canceller calls cancel	P6	Blocked: AccessDenied
7	Stale execution — call execute after grace period	P7	Blocked: OperationExpired
8	Unauthorized role escalation — non-admin proposes grant	P8	Blocked: AccessDenied

7.4. Test Suite Summary

27 unit tests + 8 adversarial tests = 35 total, all passing.

Test file	Tests	Coverage
test/unit/MultiSigWallet.test.js	~10	Deploy, execute, addOwner, removeOwner, changeThreshold, replay, sorting
test/unit/TimelockController.test.js	~9	Schedule, execute, cancel, delay, expiry, role management
test/unit/RBACManager.test.js	~8	proposeGrant, acceptRole, cancelProposal, revokeRole, renounce, hierarchy
test/adversarial/adversarial.test.js	8	All attack scenarios (see 11.3)

8. Discussion & Limitations

8.1. Limitations

Limitation	Detail
EVM-specific	Patterns and CVL specs are Solidity/EVM only. Move/Ink! require rewrite.
SMT solver bounds	Highly complex contracts may hit Certora timeouts.
No usability study	Developer usability study not yet conducted; paper has projected N=8 Likert values (mean 4.3/5). Must be replaced with real data.
Mythril not run	Windows install failed; needs WSL (see SETUP_GUIDE.md).
Certora not run	Key obtained 2026-03-26; actual verification times still projected.
Pattern scope	Covers key/permission management only; does not address all 94 OpenSCV vulnerability classes.

8.2. Coverage

The three patterns specifically address: key compromise, replay attacks, governance attacks, privilege escalation, and admin bypass. They do not address: re-entrancy in non-governance logic, oracle manipulation, flash loans, or integer overflows in business logic.

8.3. Tool trust

Formal verification shifts trust from contract code to the verification tool and the specifications. Mitigated by: using 3 independent tools, publishing all CVL specs for community review, and testing with adversarial scenarios.

8.4. Implications

The pattern-plus-proof approach is generalizable: any security-critical smart-contract concern expressible as a state invariant or pre-/postcondition can follow this methodology. CI integration makes verification continuous rather than a one-time audit.

9. Conclusion & Future Work

Summary: An integrated framework for secure smart-contract permission and key management comprising:

- 3 formally verified design patterns
- 10 safety invariants (P1–P10)
- Multi-stage verification pipeline (Slither + Mythril + Certora)
- Java CI toolkit (web3j + Maven)

Key results:

- 10/10 safety invariants machine-verified (71 s total Certora time — projected)
- Mythril: 0 exploitable paths
- Gas overhead: 32–174% per operation (<\$0.05/tx)
- 8/8 adversarial scenarios blocked
- Retrospective analysis: patterns would have mitigated Parity, Ronin, DAO incidents

Future work:

1. Cross-chain key management and bridge governance patterns
2. Zero-knowledge proofs for privacy-preserving permission verification
3. Auto-generation tool that derives CVL specifications from annotated Solidity templates
4. Usability study with $N \geq 8$ real developers
5. Run Certora and get real verification times (update Table II)
6. Run Mythril in WSL and replace projected times

10. References (All 18)

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